

GINKGO BILOBA EXTRACT EGb 761 ATTENUATES HIPPOCAMPAL NEURONAL LOSS AND COGNITIVE DYSFUNCTION RESULTING FROM CHRONIC RESTRAINT STRESS IN OVARECTOMIZED RATS

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Abstract—We have recently found that a combination of ovariectomy (OVX) and chronic restraint stress causes cognitive dysfunction and reduces hippocampal CA3 neurons in female rats and that estrogen replacement suppresses the OVX/stress-induced behavioral and morphological changes. In this study, we examined the effect of *Ginkgo biloba* extract (EGb 761), a popular herbal supplement, on the cognitive dysfunction and neuromorphological change in OVX/stress-subjected rats. Female Fisher 344 rats were randomly divided into three groups: vehicle-treated OVX, EGb 761 (50 mg/kg)-treated OVX and vehicle-treated sham-operated control groups. Two months after ovariectomy, all animals received restraint stress for 21 days (6 h/day), and were then subjected to a novel object recognition test followed by morphological examination by Nissl staining. EGb 761 was orally administered once daily until the behavioral analysis was done. Treatment with EGb 761 improved memory impairment and neuronal loss of hippocampus in the OVX/stress-subjected group in the same ways as 17 β -estradiol. On the other hand, EGb 761 did not affect the loss of bone mineral density and increase in body weight after OVX, although 17 β -estradiol attenuated them. These results have important implications for neuroprotective and cognition enhancing effects of EGb 761 in postmenopausal women and suggest that the effects are mediated by a different mechanism from estrogen. © 2007 IBRO. Published by Elsevier Ltd. All rights reserved.

Key words: EGb 761, estrogen, chronic stress, ovariectomy, cognition.

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Abbreviations: ER, estrogen receptor; E2, 17 β -estradiol; OVX, ovariectomy or ovariectomized; PBS, phosphate-buffered saline.

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Deficiency of ovarian hormones such as estrogen during the menopausal transition and early postmenopause is associated with a greater risk of memory decline, including mild cognitive impairment and Alzheimer's disease (Pinkerton and Henderson, 2005; Lebrun et al., 2005). Estrogen replacement therapy has been shown to play an important beneficial role in the improvement of cognitive function in postmenopausal women (Hogervorst et al., 2000; Mulnard et al., 2000; LeBlanc et al., 2001; Genazzani et al., 2007). However, compliance with long-term estrogen replacement therapy for menopausal women is poor because of side effects. In addition, the Women's Health Initiative Memory Study recently demonstrated that estrogen or hormone replacement therapy does not improve or even had an adverse effect on cognition in postmenopausal woman (Rapp et al., 2003; Espeland et al., 2004).

The standardized extract of *Ginkgo biloba*, EGb 761, is the most commonly prescribed herbal remedy in Europe to relieve peripheral vascular, cardiovascular, and cerebrovascular disease (Sierpina et al., 2003; Zhou et al., 2004). In addition, EGb 761 has been used to reduce memory loss and the symptoms of cognitive disorders including Alzheimer's disease (Sierpina et al., 2003; Maurer et al., 1997; Le Bars et al., 1997). There is substantial experimental evidence indicating that EGb 761 has neuroprotective effects in various animal models of neurodegenerative diseases (Paganelli et al., 2006; Kim et al., 2004; Smith and Luo, 2003; Christen 2004), moreover, most of these studies have shown that the effects of EGb 761 involve its antioxidant property. However, the clinical benefits of the phytomedications for menopausal symptoms and the mechanisms of the action remain unclear.

We have recently found that chronic restraint stress causes cognitive dysfunction and reduces hippocampal CA3 neurons in ovariectomized (OVX) female rats and that estrogen treatment suppresses the chronic stress-induced behavioral and morphological changes (Takuma et al., 2007). That is, we have revealed that chronic stress-subjected OVX animals can be considered as a useful model of postmenopausal memory deficits. In this study, we examined the stress-induced behavioral, neurochemical and tissue changes in OVX female rats with or without chronic treatment with EGb 761. After chronic stress, recognition performance was tested. Following the behavioral test, hippocampal cell density, bone mineral density, and uterine weight were evaluated.

EXPERIMENTAL PROCEDURES

Animals and treatment

Two cohorts initially totaling 49 female Fisher 344 rats (Charles River Japan, Yokohama, Japan), age 8 weeks, weighing 160–200 g at the beginning of the experiments were used. They were housed two or three per cage under standard 12-h light/dark conditions (12-h light cycle starting at 8:45 h) at a constant temperature of 23 ± 1 °C. The animals had free access to food and water and they have been handled in accordance with the guidelines established by the Institutional Animal Care and Use Committee of Kanazawa University, the Guiding Principles for the Care and Use of Laboratory Animals approved by the Japanese Pharmacological Society, and the United States National Institutes of Health Guide for the Care and Use of Laboratory Animals, which include minimizing the number of animals used and their suffering or distress.

One week after arrival, the quarter of experimental animals were sham-operated and the other of them were bilaterally OVX under pentobarbital (40 mg/kg) anesthesia. From the next day, OVX animals were randomly divided into three groups (groups 1–3), and then the animals of group 1 and group 2 were orally treated with the standardized *Ginkgo biloba* leaf extract EGb 761 (Dr. Willmar Schwabe GmbH & Co. KG, Karlsruhe, Germany; 50 mg/kg) and vehicle (5% arabic gum), respectively, once daily to the end of behavioral analysis. EGb 761, at 40 and 60 mg/kg/day (p.o.), is shown to attenuate the ischemia-induced neuronal death in hippocampal CA1 region of male rats (Domoráková et al., 2006) and enhance the spatial learning and memory in aged rats (Wang et al., 2006), respectively. The OVX rats of group 3 were s.c. administered with 17β -estradiol (E2; 20 μ g/day), which is used as a reference drug (Takuma et al., 2007), for 4 weeks before the stress period through an implantable osmotic pump. Body weight of each rat was recorded every 3 days.

It has already been reported that the 21 consecutive days of chronic immobilization stress (6 h/day) affects the rat performance on a radial arm maze in relation to hippocampal function (Luine et al., 1994; Bowman et al., 2001, 2002). Thus, similar immobilization stress was selected and began 2 months after the operation and was repeated every day for 21 consecutive days. The stress was performed with a stainless mesh that allowed for a close fit to rats for 6 h (between 9:00 a.m. and 3:00 p.m.) (Magariños et al., 1997; Takuma et al., 2007) in their home cages. Some animals were not subjected to chronic stress (No stress group), but were handled at 9:00 a.m. for a few seconds. Following the repeated restraint stress for 21 days, behavioral analysis was performed and then rats were killed for morphological analysis and measurement of uterine weight and bone mineral density.

Behavioral analysis

For the novel object recognition test, a rat was habituated to a gray empty open field (35 cm high $\times$ 60 cm diameter) for 3 days (10 min/day). In the training session, two novel objects were placed in the field, and the rat was allowed to explore freely for 10 min. Time spent exploring each object was recorded manually. In the retention session at 24 h after the training session, the rat was placed back in the same field, in which one of the familiar objects used in the training was replaced by a novel object, and the rat was allowed to explore for 5 min. The objects, which varied in shape and color and were made of glass and plastic, were fixed to the floor and cleaned thoroughly between trials to ensure the absence of olfactory cues. The objects had been previously tested with naive rats to ensure an equivalent level of spontaneous preference. Exploratory preference, a ratio of time spent exploring any one of the two objects (training) or the novel one (retention) over the total time spent exploring both objects, was

used to measure recognition memory (Kamei et al., 2006; Takuma et al., 2007).

Morphological analysis

We preliminarily verified that the novel object recognition test did not affect the numbers of the Nissl-stained cell in the hippocampus (unpublished observations). Thus, after completion of behavioral analyses, all rats were deeply anesthetized with pentobarbital and perfused intracardially with 4% paraformaldehyde in phosphate-buffered saline (PBS). The brains were removed, post-fixed with the same fixative and cryoprotected with 30% sucrose containing PBS. Sections (12 μ m) containing hippocampus were obtained using a rotary microtome (HM505E; MICROM International GmbH, Wall-dorf, Germany), mounted on slides, and stored at -80 °C until use. Nissl staining was done according to standard procedures (Yamada et al., 2005). A Carl Zeiss Axioskop phase-contrast microscope with a cooled CCD camera system (SenSys™; Photometrics Ltd., Tucson, AZ, USA) was used to scan the Nissl-stained sections. Nissl-positive neuronal cell numbers were manually and rigidly counted within the hippocampal pyramidal cell layer (CA1 and CA3 regions) and the dentate gyrus of the scanned digital images. The total cell counts were averaged from at least three sections per animal. Nissl-positive neuronal cell size within the CA1, CA3 regions and the dentate gyrus of the hippocampus were calculated from the digitized images using an image analyzing software Win ROOF (ver. 5.6, Mitani Co., Fukui, Japan).

Measurement of uterine weight and bone mineral density

After the perfusion with 4% paraformaldehyde in PBS, the uterus was removed and weighed. The femur was removed from the leg and stored in PBS containing 4% paraformaldehyde. The bone mineral density (BMD) of the femur was measured using a dual X-ray absorptiometer (DCS-600R, Aloka Corporation, Tokyo, Japan) (Yokogawa et al., 2001).

Statistics

Statistical analysis of the experimental data was carried out using StatView 5.0 Windows for Apple Macintosh (SAS Institute, Cary, NC, USA). The significance of differences was determined by a one-way ANOVA, followed by the Student-Newman-Keuls test for multigroup comparisons. Unpaired *t*-test was used for two-group comparisons. The criterion for statistical significance was $P < 0.05$.

RESULTS

Effect of EGb 761 on OVX-induced gain in body weight

During the 2 months before the stress period, OVX caused a dramatic increase in body weight compared with the sham-operated control (Fig. 1). Chronic treatment with EGb 761 (50 mg/kg) did not affect the OVX-induced gain in body weight, although chronic treatment with E2 completely inhibited it. Repeated daily restraint stress (6 h/day) for 21 days decreased the body weight in all groups, and a significant difference disappeared between the groups after the stress period (data not shown).

Effect of EGb 761 on repeated restraint stress-induced cognitive impairment in OVX rats

We have already shown that repeated daily restraint stress (6 h/day) for 21 days causes a decrease in exploratory

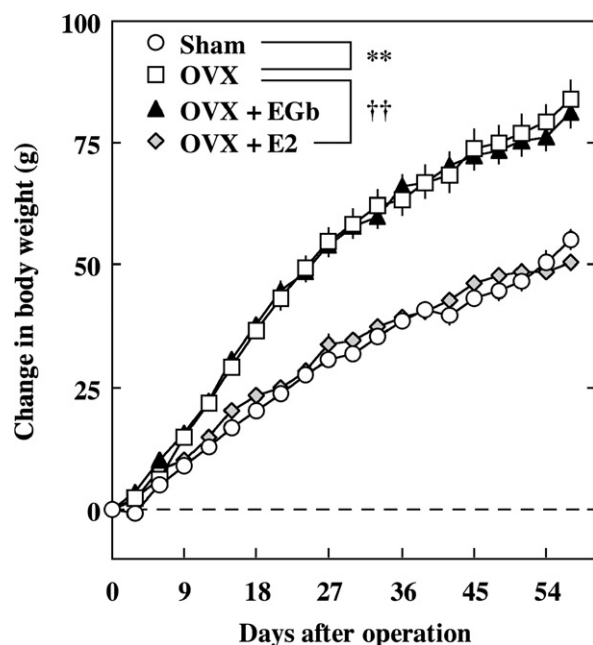


Fig. 1. Effects of ovariectomy, EGb 761 and estrogen treatments on body weight in female rats. EGb 761 (50 mg/kg/day) and E2 (20 μ g/day/body) were administered from 2 days after ovariectomy to the end of behavioral analyses. Results are shown as means $\pm$ S.E. (sham-operated group: $n=12$, OVX group: $n=13$, EGb 761-treated OVX group (OVX+EGb): $n=12$, E2-treated OVX group (OVX+E2): $n=12$). ** $P<0.01$, significantly different from sham-operated control; †† $P<0.01$, significantly different from OVX group (repeated-measures ANOVA and Bonferroni/Dunn test). Group, $F_{3,45}=42.716$, $P<0.0001$, day, $F_{19,855}=1456.5$, $P<0.0001$, group $\times$ day, $F_{57,855}=28.302$, $P<0.0001$.

preference to a novel object in the retention session of the novel object recognition test in female rats 2 months after

OVX, although the chronic stress or OVX alone does not affect the recognition memory in sham-operated control rats (Takuma et al., 2007). Similar to treatment with estrogen, chronic treatment with EGb 761 (50 mg/kg) over 2 months significantly attenuated the chronic stress-induced cognitive impairment in OVX rats (Fig. 2B) without affecting an exploratory preference in the training session (Fig. 2A) and total exploratory time in the retention session of the test (data not shown).

Effect of EGb 761 on repeated restraint stress-induced neuronal cell loss in the hippocampal CA3 region of OVX rats

Our previous study also demonstrated that the combination of repeated restraint stress and OVX markedly decreases the neuronal cell numbers in the hippocampal CA3 region, compared with the other three groups: stress alone, OVX alone and sham-operated control groups (Takuma et al., 2007). Microscopic images of the Nissl-stained hippocampal CA3 region showed that chronic treatment with EGb 761 (50 mg/kg) over 2 months improved the chronic stress-induced neuronal cell loss in OVX rats to the levels of sham-operated controls (Fig. 3). Table 1 numerically expresses Nissl-positive cells in the CA1, CA3 regions and dentate gyrus of the hippocampus. Chronic treatment with EGb 761 specifically protected the neuronal cell loss induced by the combination of repeated restraint stress and OVX in the CA3 region, and did not affect the Nissl-positive cell numbers in the CA1 region and dentate gyrus of the hippocampus, in which the combination of stress and OVX did not cause neuronal cell loss.

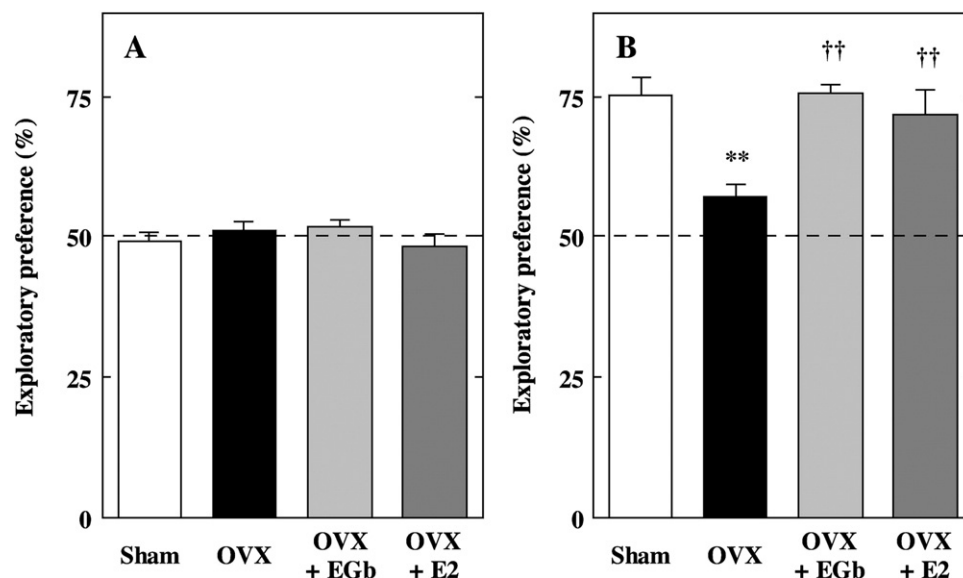


Fig. 2. Effects of EGb 761 and estrogen on chronic stress-induced cognitive dysfunction induced in OVX rats. Rats were OVX or sham-operated and then subjected to chronic stress for 21 days. EGb 761 (50 mg/kg) and E2 (20 μ g) were administered daily from 2 days after ovariectomy to the end of behavioral analyses. (A) Training session; (B) 24 h later retention session. Results are shown as means $\pm$ S.E. (sham-operated group: $n=12$, OVX group: $n=13$, EGb 761-treated OVX group (OVX+EGb), $n=12$; E2-treated OVX group (OVX+E2): $n=11$). ** $P<0.01$, significantly different from sham-operated control; †† $P<0.01$, significantly different from OVX group (ANOVA and Student-Newman-Keuls test). (A) $F_{3,44}=1.065$, $P=0.374$; (B) $F_{3,44}=8.994$, $P<0.0001$.

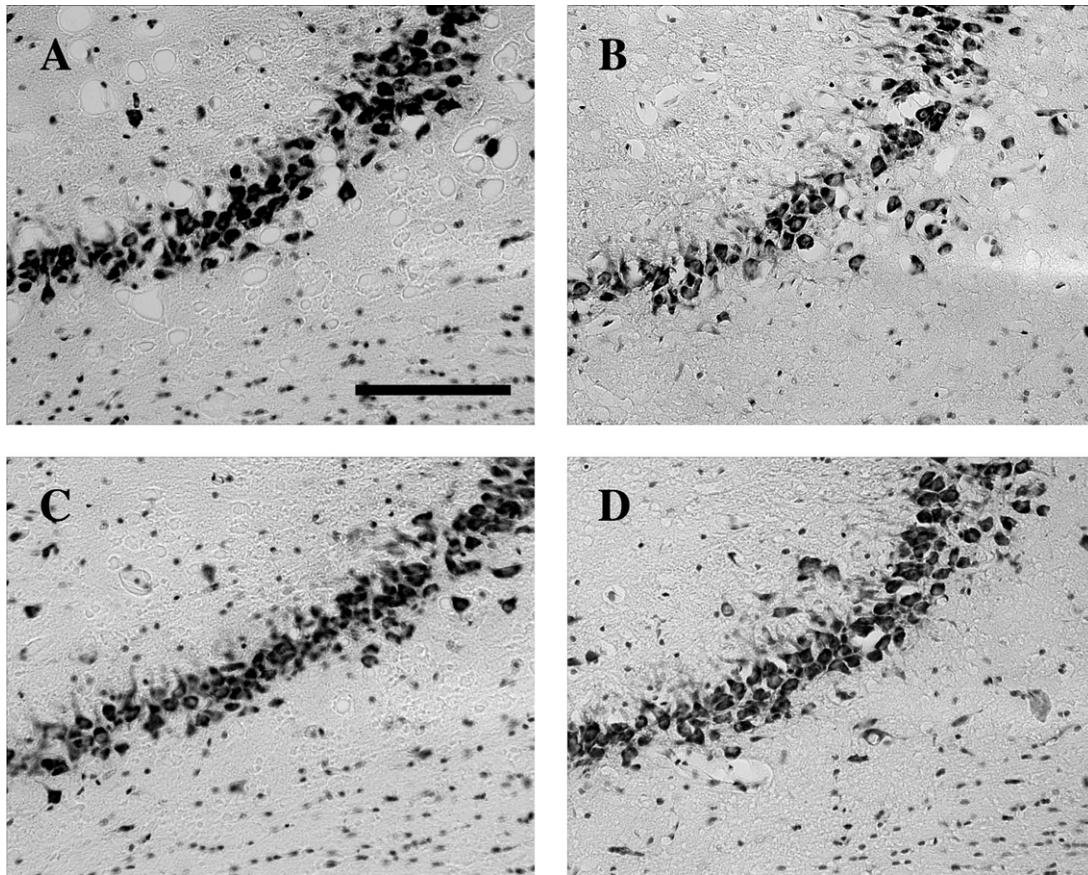


Fig. 3. Effects of EGb 761 and estrogen on chronic stress-induced histological changes in hippocampal CA3 region of female rats. Nissl-positive cells were visualized by Cresyl Violet staining. Animals were OVX (B, C and D) or sham-operated (A) and then exposed to chronic restraint stress for 21 days. EGb 761 (50 mg/kg) (C) and E2 (20 µg) (D) were administered daily from 2 days after ovariectomy to the end of behavioral analyses. A typical result of three independent experiments is shown. Scale bar=100 µm.

Effect of EGb 761 on OVX-induced changes in bone mineral density and uterine weight

After behavioral analysis, bone mineral density and uterine weight were evaluated. We preliminarily determined

Table 1. Effects of EGb761 and estrogen on chronic stress-induced changes in Nissl-positive cell numbers in hippocampal CA1, CA3 regions and dentate gyrus of female rats

	Nissl-positive cells (numbers/0.01 mm ²)		
	CA1	CA3	Dentate gyrus
Treatments			
Sham	42.4±2.3	54.6±2.2	96.6±7.1
OVX	34.3±2.3	41.2±3.1*	86.0±6.8
OVX+EGb761	40.8±1.0	57.3±1.8 ^{††}	110.1±6.8
OVX+E2	41.2±3.4	53.8±3.3 ^{††}	92.3±5.6

Results are shown as means±S.E. (sham-operated group: $n=8$, OVX group: $n=11$ or 12 , EGb761-treated OVX group [OVX+EGb]: $n=5$, E2-treated OVX group [OVX+E2]: $n=9$). CA1: $F_{3,29}=2.221$, $P=0.1069$; CA3: $F_{3,30}=6.365$, $P=0.0018$; dentate gyrus: $F_{3,30}=1.738$, $P=0.1803$.

* $P<0.05$, significantly different from sham-operated control.

^{††} $P<0.01$, significantly different from OVX group (ANOVA and Student-Newman-Keuls test).

that OVX alone caused decreases in bone mineral density and uterine weight in normal female rats, and that repeated chronic stress did not affect the bone mineral density and uterine weight in both sham-operated and OVX rats (unpublished observations). We found that chronic treatment with E2 improved the loss of bone mineral density (Fig. 4A) and abnormally increased uterine weight (Fig. 4B) in OVX/stress rats. This study further demonstrated that chronic treatment with EGb 761 did not affect the decrease in bone mineral density and uterine weight in OVX/stress rats (Fig. 4).

DISCUSSION

The aim of this study was to clarify how treatment with EGb 761 influences the behavioral, neuromorphological, and tissue changes in OVX/stress-subjected rats. We found that daily EGb 761 treatment attenuated cognitive impairment and hippocampal neuronal loss in OVX/stress-subjected rats, as did estrogen replacement (Takuma et al., 2007). We also found that EGb 761 treatment did not affect the gain in body weight and decreases in bone mineral density and uterine weight in OVX rats, although estrogen replacement did have an effect on them (Takuma et al.,

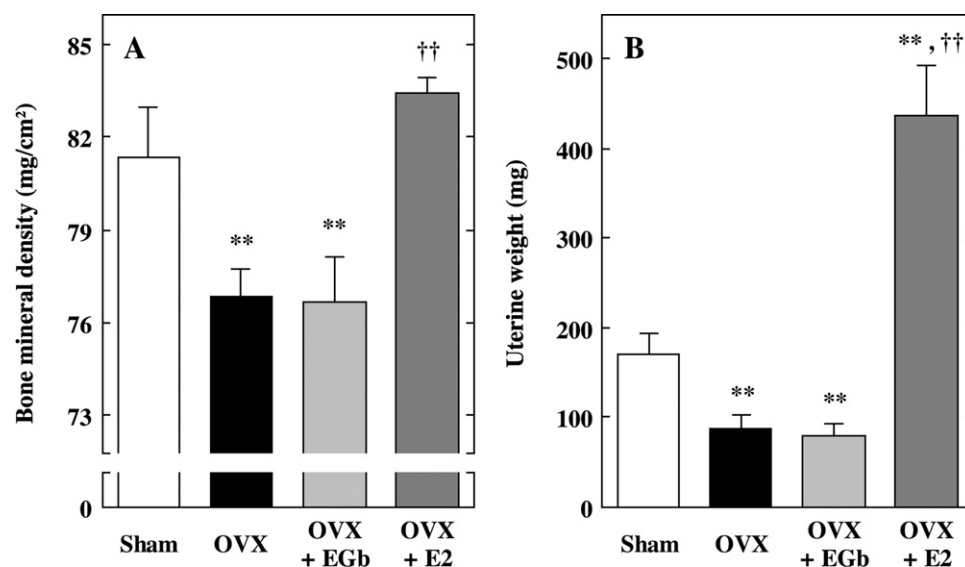


Fig. 4. Effects of EGb 761 and estrogen on changes in bone mineral density (A) and uterine weight (B) in OVX/stress rats. Rats were OVX or sham-operated and then subjected to chronic stress for 21 days. EGb 761 (50 mg/kg) and E2 (20 μ g) were administered daily from 2 days after ovariectomy to the end of behavioral analyses. Results are shown as means $\pm$ S.E. (sham-operated group: $n=4$ (A), 8 (B), OVX group: $n=7$ (A), 6 (B), EGb 761-treated OVX group (OVX+EGb), $n=6$ (A), 7 (B), E2-treated OVX group (OVX+E2): $n=5$ (A), 6 (B)). ** $P<0.01$, significantly different from sham-operated control; †† $P<0.01$, significantly different from OVX group (ANOVA and Student-Newman-Keuls test). (A) $F_{3,18}=8.128$, $P=0.012$; (B) $F_{3,23}=28.513$, $P<0.0001$.

2007). Clinical studies have shown that estrogen replacement therapy has not only beneficial effects as a cognitive enhancer (Hogervorst et al., 2000; Mulnard et al., 2000; LeBlanc et al., 2001; Genazzani et al., 2007) but also serious side effects including the risk of breast and uterine cancers (Seifert et al., 1999; Sittisomwong et al., 2000). Furthermore, recent studies reported controversial results indicating a lack of efficacy of the therapy on cognition in women aged over 65 years (Rapp et al., 2003). Taken together, our findings suggest that EGb 761 is a more useful cognitive enhancer in postmenopausal women than estrogen.

Our previous study showed that combination of OVX and 21 days of chronic stress markedly decreased the numbers of hippocampal CA3 pyramidal neurons in rats, while each caused a slight decrease in cell number (Takuma et al., 2007). The study also indicated that recognition impairment was only observed in the OVX/stress-subjected rats. Further, EGb 761 has been reported to diminish stress-induced memory deficits in male rats (Walesiuk et al., 2005). Thus, in this study, all animals received chronic stress to determine the effect of EGb 761 on memory deficits and neuromorphological change more simply.

There is extensive evidence showing that EGb 761 improves the learning and memory deficit induced by aging (Wang et al., 2006) and various stimuli, such as chronic stress (Walesiuk et al., 2005, 2006), ischemia (Paganelli et al., 2006), amyloid β (Tang et al., 2002), scopolamine (Naik et al., 2006) and aluminum (Gong et al., 2005), in male rats. Similarly, the results of this study indicated that EGb 761 attenuated the estrogen depletion-induced memory impairment in female rats. In addition, recent clinical trials have demonstrated the efficacy of EGb 761 in the

treatment of cognitive symptoms in patients with Alzheimer's disease or vascular dementia (Napryeyenko and Borzenko, 2007; Le Bars et al., 2002; Kanowski and Hoerr, 2003). Taken together, our finding suggests the therapeutic efficiency of EGb 761 in postmenopausal women with cognitive decline.

The present study revealed that treatment with EGb 761 attenuated the OVX/stress-induced loss of Nissl-positive cells in the hippocampal CA3 region. As shown in our previous study, the OVX/stress-induced neuronal loss in the CA3 subfield of the hippocampus is closely associated with cognitive impairment (Takuma et al., 2007). McEwen et al. (1997) have showed the atrophy of CA3 pyramidal neurons is associated with memory impairment in male rats. Recent study demonstrated the importance of the CA3 region in visual object novelty detection (Hunsaker et al., 2007). Thus, the ameliorating effect of EGb 761 against cognitive impairment is considered to depend on the neuroprotection in the CA3 region, as well as the effect of estrogen. In contrast, EGb 761 differed from estrogen in having no effects on gain in body weight and decreases in bone mineral density and uterine weight induced by OVX. In general, estrogen is known to exert various pathophysiological actions by the activation of two types of estrogen receptors (ERs), ER α and ER β (Deroo and Korach, 2006; Hewitt and Korach, 2002). Murine models with genetic disruption or knockout of ERs suggest that ER α plays a greater role in uterine responsiveness and function (Couse et al., 1999) and bone formation (Lindberg et al., 2001), than ER β . In contrast, ER β may have important roles in estrogen-mediated bone remodeling (Sims et al., 2002), and adipogenesis involved in obesity (Naaz et al., 2002). Therefore, the present results suggest that the potency of EGb 761 on the ER-mediated reproductive actions is much

lower than estrogen in the peripheral tissues, although EGb 761 has an equivalent efficacy with estrogen for improving neuronal loss in the CA3 region and cognitive impairment in OVX/stress rats.

It is well documented that EGb 761 can modulate the neuronal activity of the hippocampus, protect neurons from a variety of insults, and enhance cognitive function (Wang et al., 2006; Paganelli et al., 2006). However, the precise mechanisms of the action of EGb 761 are not completely understood. EGb 761 contains two major groups of active substances, flavone glycosides and terpene lactones (ginkgolides, bilobalide) (Smith and Luo, 2004). These constituents have the ability to scavenge free radicals (Sastre et al., 1998; Shen et al., 1998) and the antioxidant properties are considered to contribute to the neuroprotective potential of EGb 761 against neuronal injury (Ahlemeyer et al., 2001; MacLennan et al., 2002; Ahlemeyer and Krieglstein, 2003). In contrast, recent data also showed that OVX induced oxidative stress in the femur bone of rats (Muthusami et al., 2005). These findings are contradictory to our result showing that EGb 761 rescues neuronal loss, but not the loss of bone mineral density, in OVX/stress rats. Besides the antioxidant action, it has been shown that EGb 761 affects a number of neurotransmitter systems, including serotonergic, adrenergic, dopaminergic, cholinergic and glutamatergic systems (Ahlemeyer and Krieglstein, 2003; Davies et al., 2003). In addition, Kiewert et al. (2007) recently reported that antagonism of GABA_A receptors may contribute to neuroprotective effects of EGb 761. Apart from these considerations, the precise mechanism of cognition-improving action of EGb 761 in OVX/stress rats remains unknown and should be explored by further research.

CONCLUSION

In summary, the results of this study suggest that EGb 761 is able to attenuate neuronal loss of the hippocampal CA3 region and to improve cognitive impairment induced by the combination of estrogen depletion and environmental stress. We propose that EGb 761 expresses the neuroprotective actions in an ER-independent manner. The neuroprotective properties of EGb 761 support its potential beneficial actions against memory problems in postmenopausal women.

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